Activity 4 Worksheet - Circularity and Decarbonisation Portfolio

Exact procedure

1. Open mock-data/mock-data.csv. Separate negative annual cost (a saving) from positive annual cost.

2. Calculate marginal abatement cost: MAC = annual_net_cost_sgd / annual_abatement_tco2e. Worked example for compressed-air optimisation: -85,000 / 420 = -S$202.38/tCO2e.

3. Rank initiatives from lowest to highest MAC, but mark dependencies and delivery time before selecting the sequence.

4. Quantify portfolio abatement by year; do not add an initiative until its dependency and start date are satisfied.

5. Compare MAC with Singapore's carbon-tax context: S$45/tCO2e for 2026-2027. Carbon tax alone is not a complete investment case.

6. Classify the sequence: no-regret (negative MAC), enabling (creates later capability), transformation (material business-model change).

7. State ecological and community effects, evidence owner and verification method for each selected initiative.

Initiative	MAC (S$/tCO2e)	Sequence class	Start	Dependency	Verified abatement	Ecological/community effect